305141
Introduction to Internet of Things

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Chapter 06
Basic of Arduino Platform

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Background of Arduino

www.arduino.cc



Background of Arduino

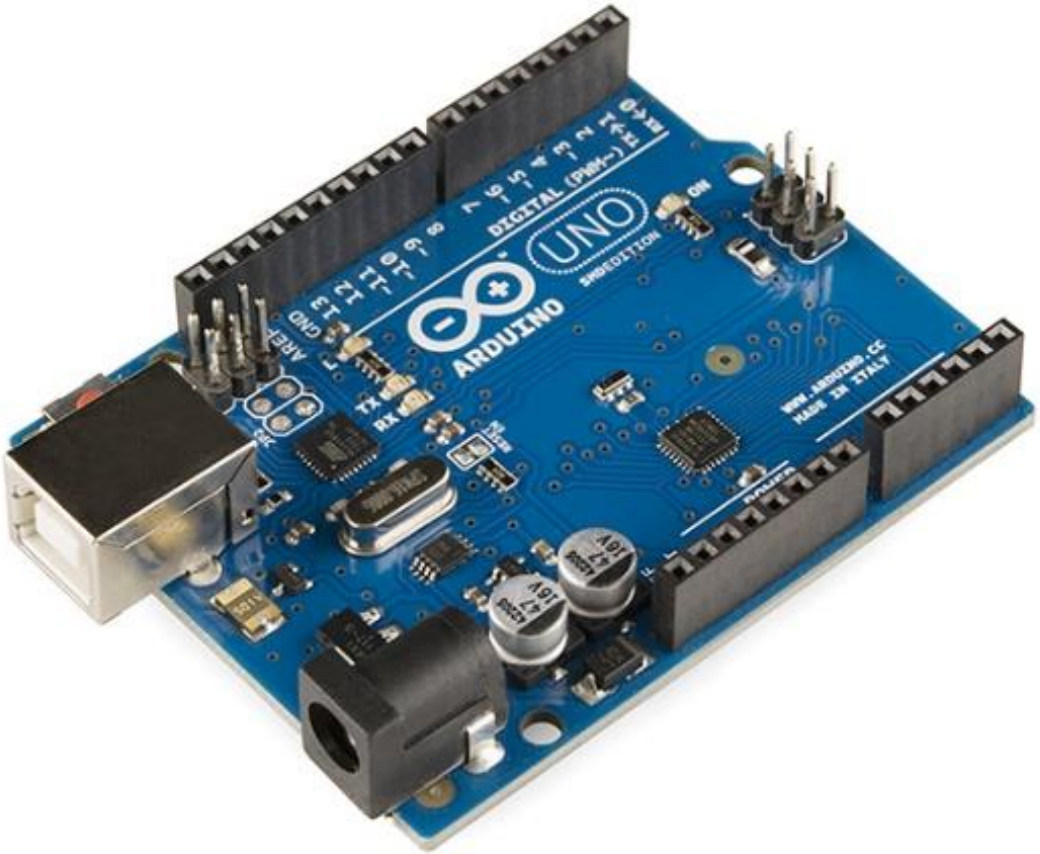
"Arduino is an open-source electronics prototyping platform based on flexible, easy-to-use hardware and software. It's intended for artists, designers, hobbyists, and anyone interested in creating interactive objects or environments." [www.arduino.cc]

Background of Arduino

- Invented in Italy, 2005 .
- By Massimo Banzi and David Cuartielles.
- For purpose, to make a small price microcontroller for students.
- Develop based on Open Source Platform (free) both Hardware and Software.
- Develop by Wiring System.

Arduino UNO R3



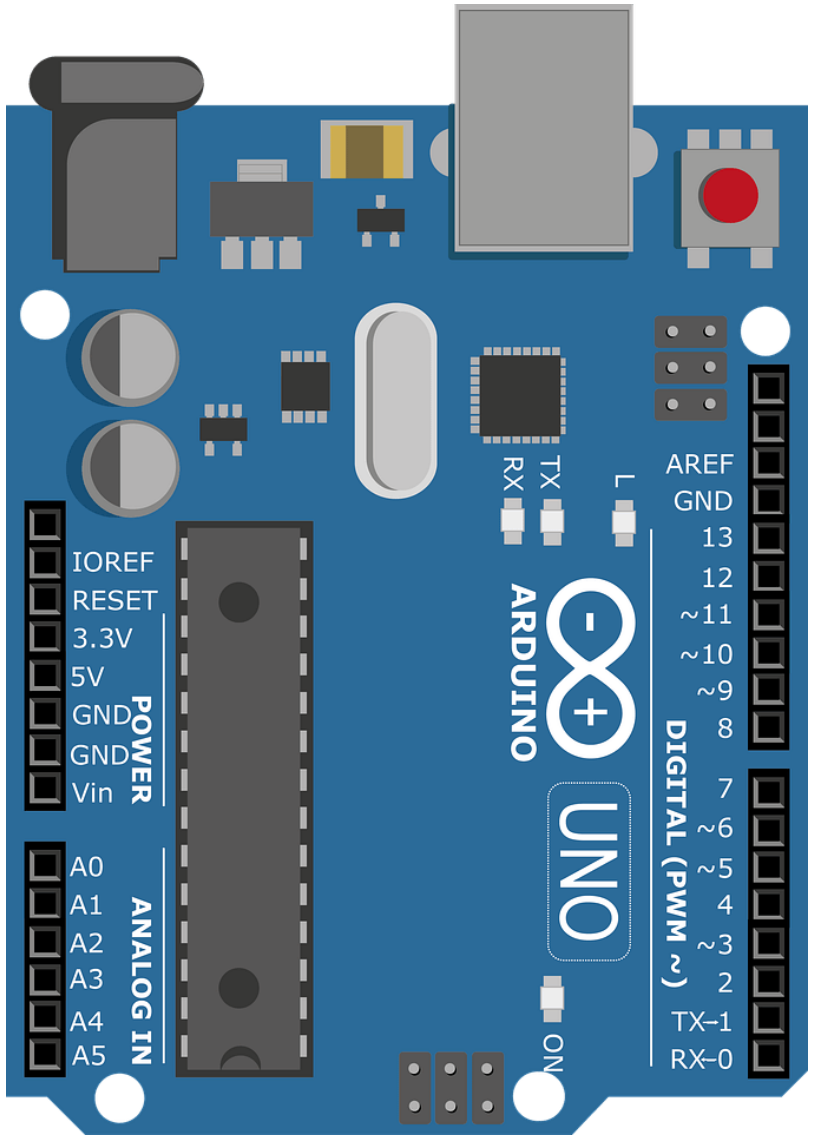


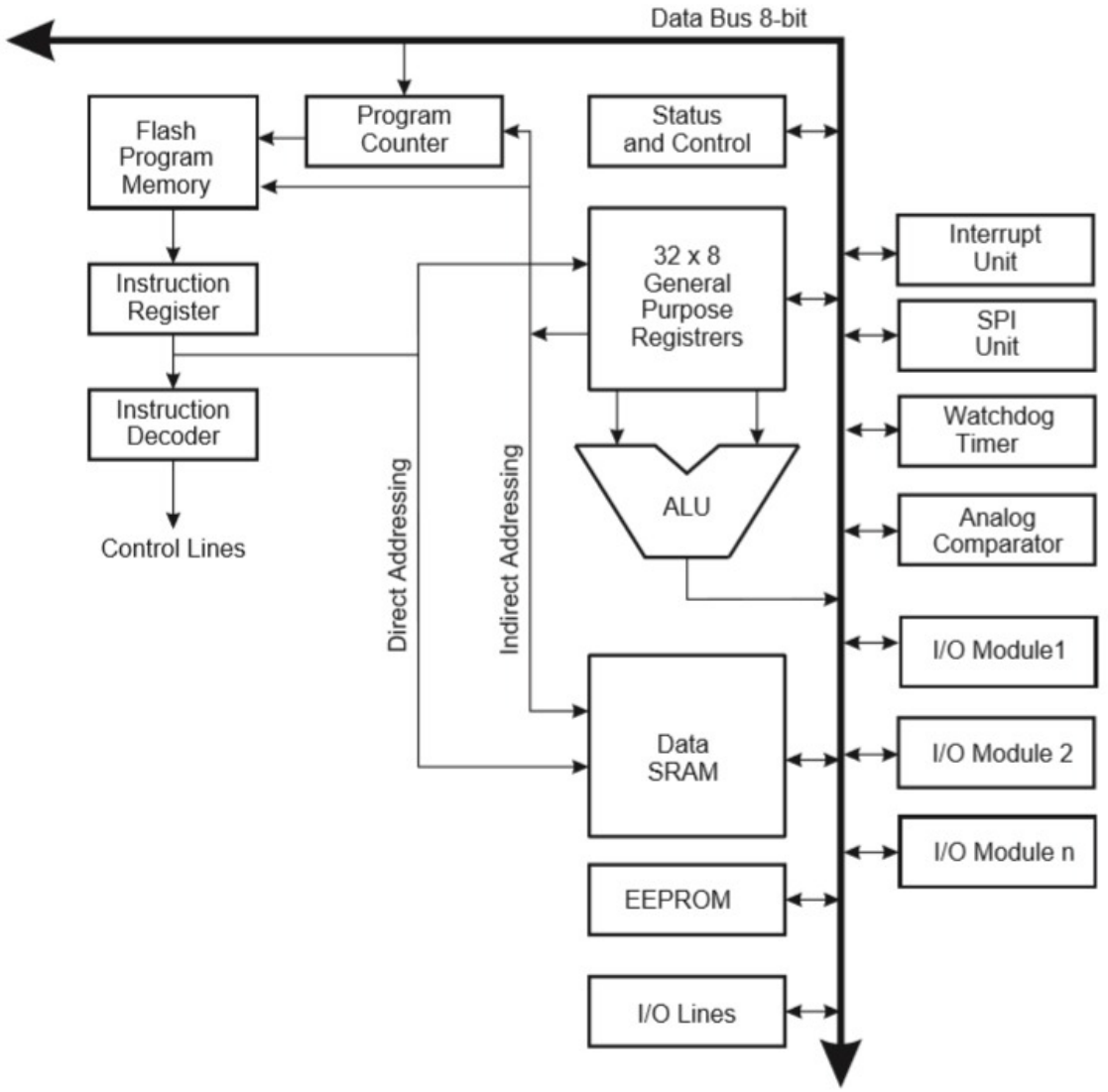
Arduino UNO R3

AVR

Microcontroller Chip	ATmega328
Voltage Supply	5V
Recommended Power Supply	7 – 12V
Limited Power Supply	6 – 20V
Digital I/O Port	14 port (6 PWM output)
Analog Input Port	6 พอร์ต
Current supply for each port	40mA
Current supply for each port 3.3V	50mA
Programmable ROM	32KB, 500B for Booloader
RAM	2KB
EEPROM	1KB
Crystal Frequency	16MHz
Size	68.6x53.4 mm
Weight	25 gram.

Arduino UNO R3





Arduino UNO R3

The basic working of CPU of ATmega328:-

1. The data is uploaded in serial via the port (being uploaded from the computer's Arduino IDE). The data is decoded and then the instructions are sent to **instruction register** and it decodes the instructions on the same clock pulse.

2. On the next clock pulse the next set of instructions are loaded in instruction register.

3. In general purpose registers the registers are of 8-bit but there are 3 16-bit registers also.

a. 8-bit registers are used to store data for normal calculations and results.

b. 16-bit registers are used to store data of timer counter in 2 different register. Eg. X-low & X-high.

Arduino UNO R3

4. EEPROM stores data permanently even if the power is cut out. Programming inside a EEPROM is slow.

5. Interrupt Unit checks whether there is an interrupt for the execution of instruction to be executed in ISR (Interrupt Service Routine).

6. Serial Peripheral Interface (SPI) is an interface bus commonly used to send data between microcontrollers and small peripherals such as Camera, Display, SD cards, etc. It uses separate clock and data lines, along with a select line to choose the device you wish to talk to.

Arduino UNO R3

7.Watchdog timer is used to detect and recover from MCU malfunctioning.

8. Analog comparator compares the input values on the positive and negative pin, when the value of positive pin is higher the output is set.

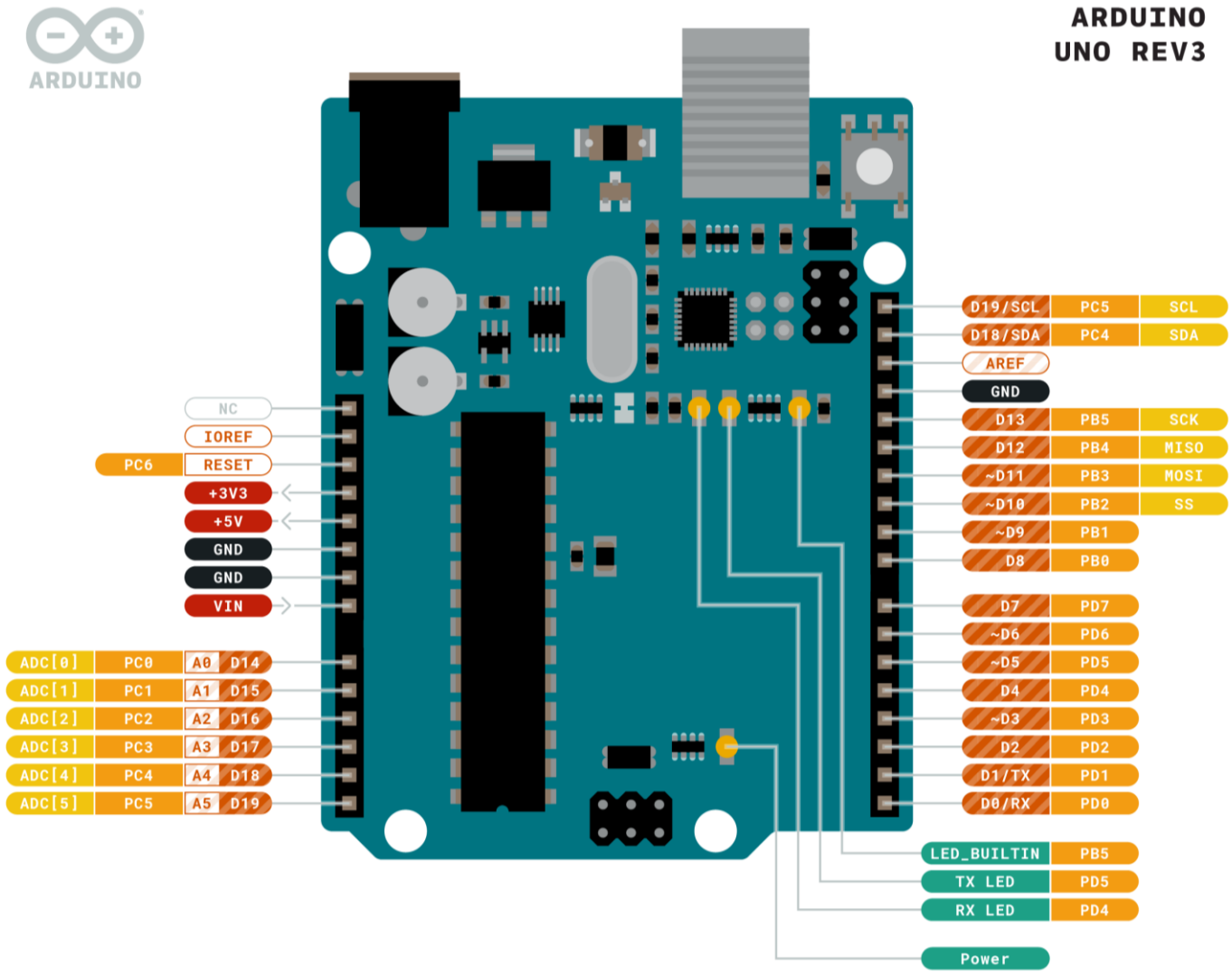
9. Status and control is used to control the flow of execution of commands by checking other blocks inside the CPU at regular intervals.

Arduino UNO R3

10. ALU (Arithmetic and Logical unit) The high performance AVR ALU operates in direct connection with all the 32 general purpose working registers. Within a single clock cycle, arithmetic operations b/w general purpose registers are executed. The ALU operations are divided into 3 main categories – arithmetic, logical and bit-function.

11. I/O pins The digital inputs and outputs (digital I/O) on the Arduino are what allow you to connect the Arduino sensors, actuators, and other ICs. Learning how to use them will allow you to use the Arduino to do some really useful things, such as reading switch inputs, lighting indicators, and controlling relay outputs.

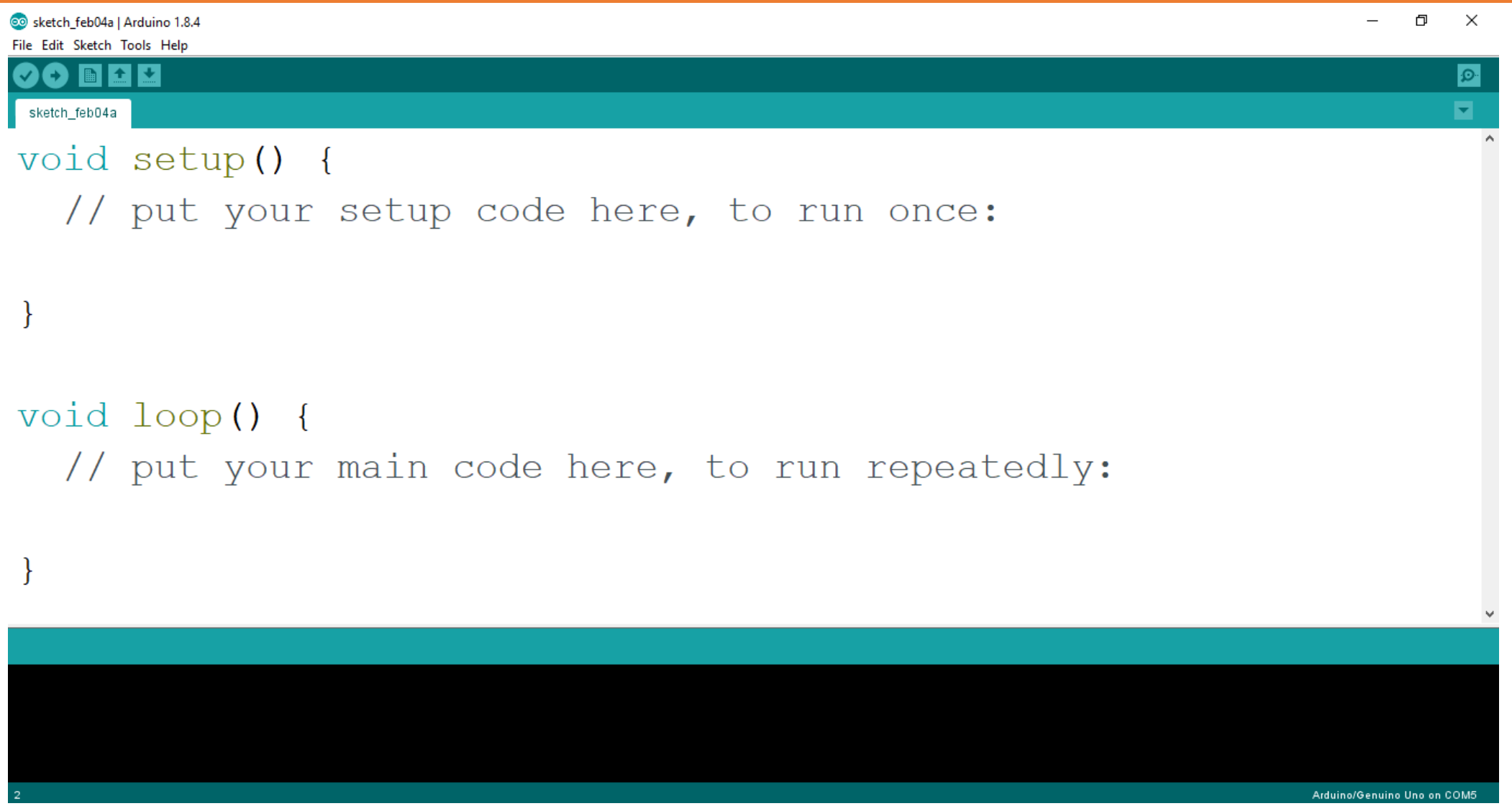
Arduino UNO R3



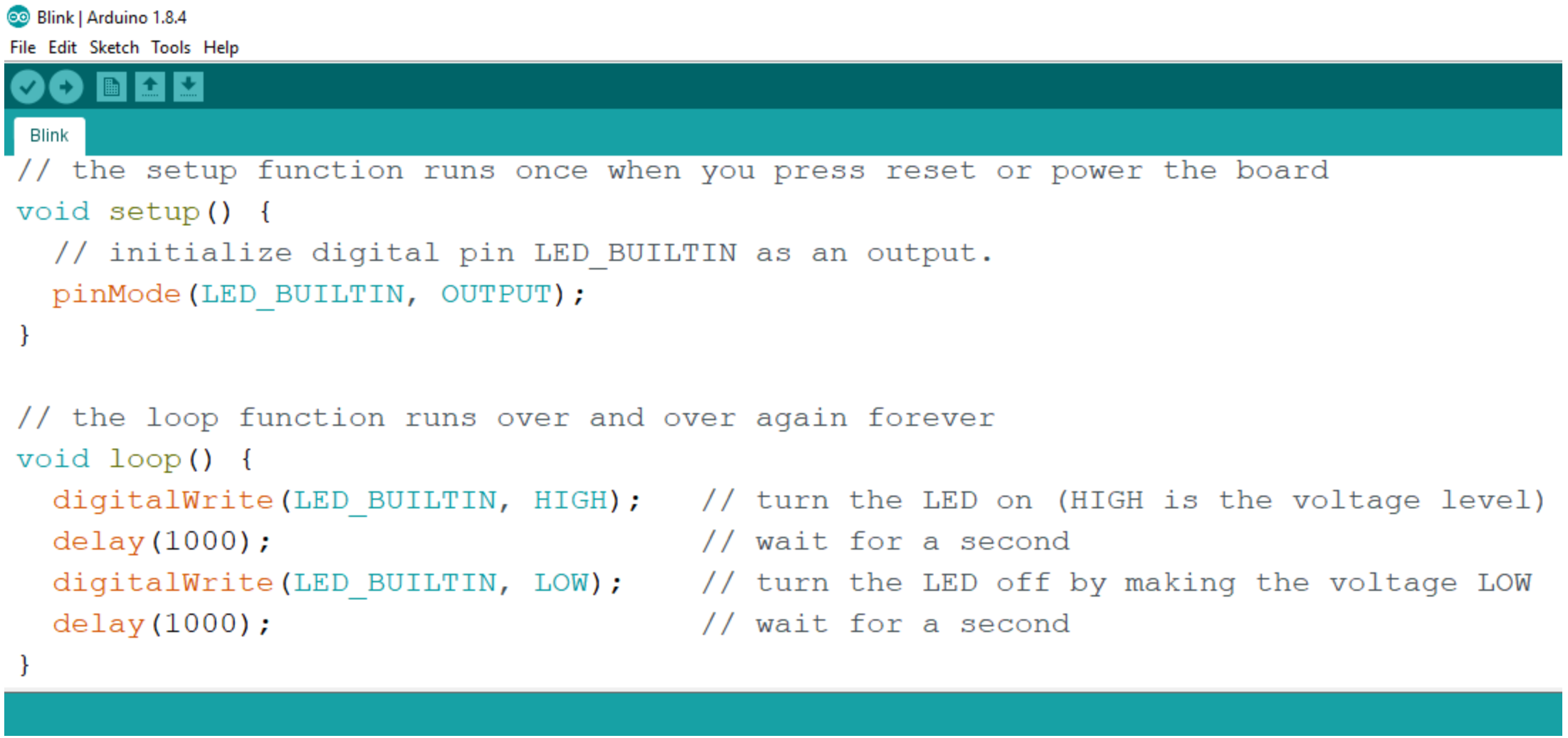
Ground
Power
LED
Internal Pin
SWD Pin
Other Pin

Digital Pin
Analog Pin
Default
Microcontroller's Port

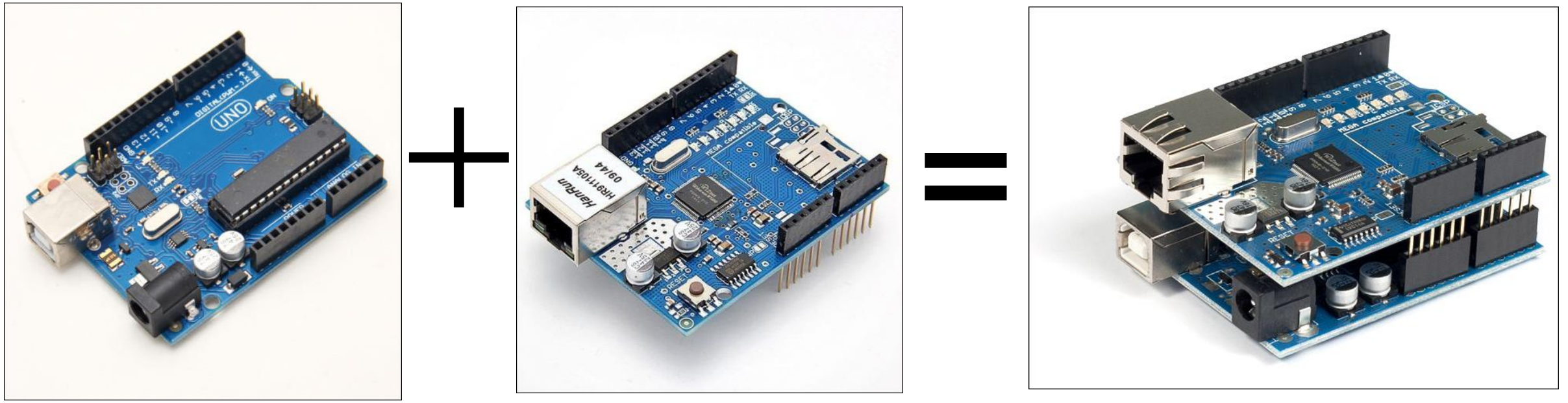
Arduino IDE


sketch_feb04a | Arduino 1.8.4
File Edit Sketch Tools Help
sketch_feb04a
void setup() {
// put your setup code here, to run once:
}
void loop() {
// put your main code here, to run repeatedly:
}
2
Arduino/Genuino Uno on COM5

Arduino IDE

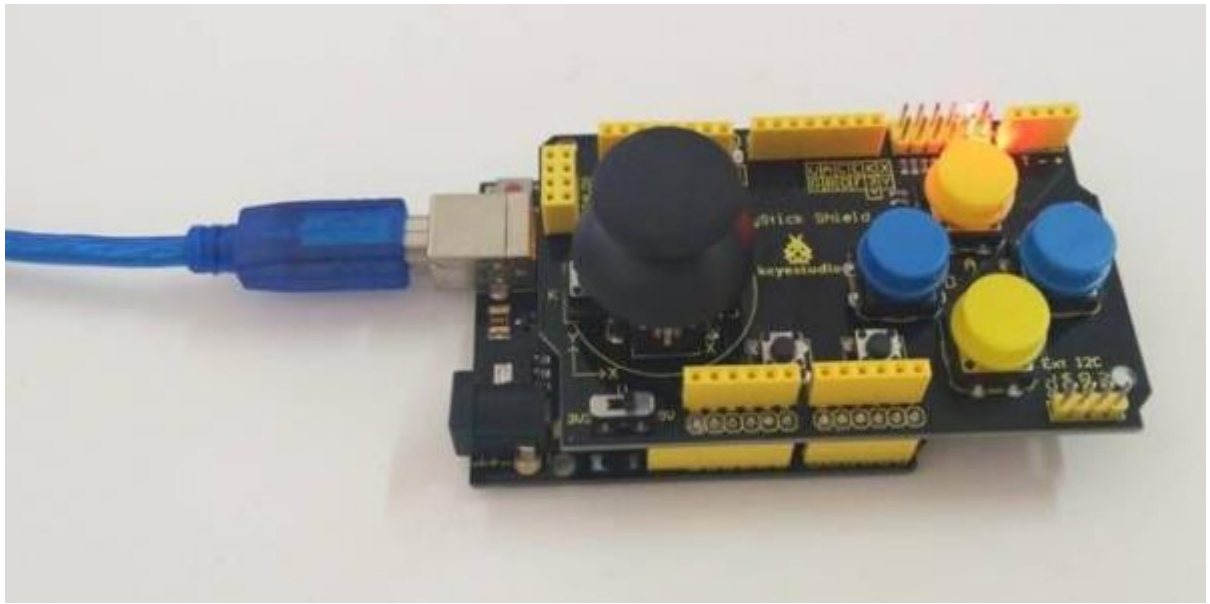

Blink | Arduino 1.8.4
File Edit Sketch Tools Help
Blink
// the setup function runs once when you press reset or power the board
void setup() {
// initialize digital pin LED_BUILTIN as an output.
pinMode(LED_BUILTIN, OUTPUT);
}
// the loop function runs over and over again forever
void loop() {
digitalWrite(LED_BUILTIN, HIGH); // turn the LED on (HIGH is the voltage level)
delay(1000); // wait for a second
digitalWrite(LED_BUILTIN, LOW); // turn the LED off by making the voltage LOW
delay(1000); // wait for a second
}

Arduino Shield

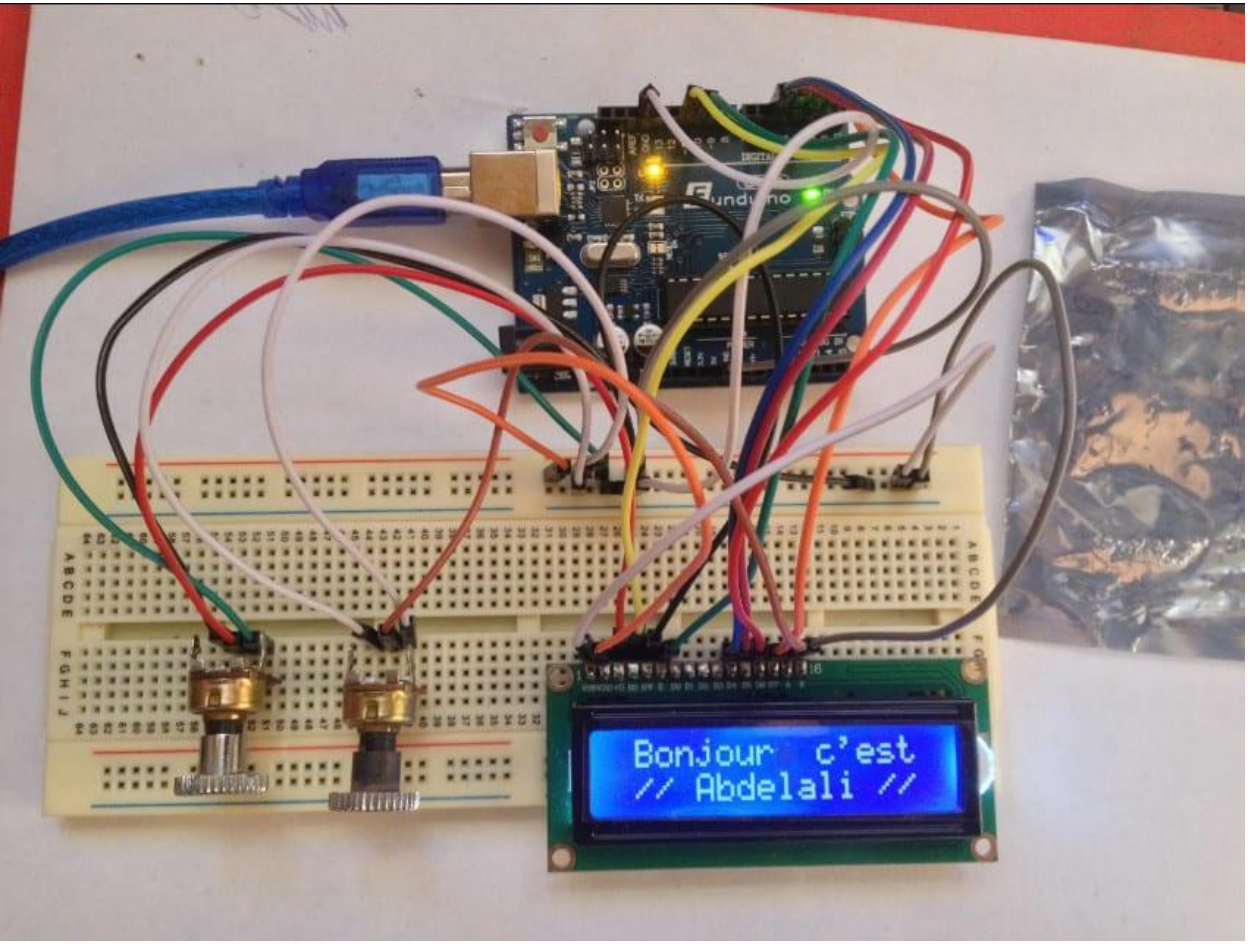


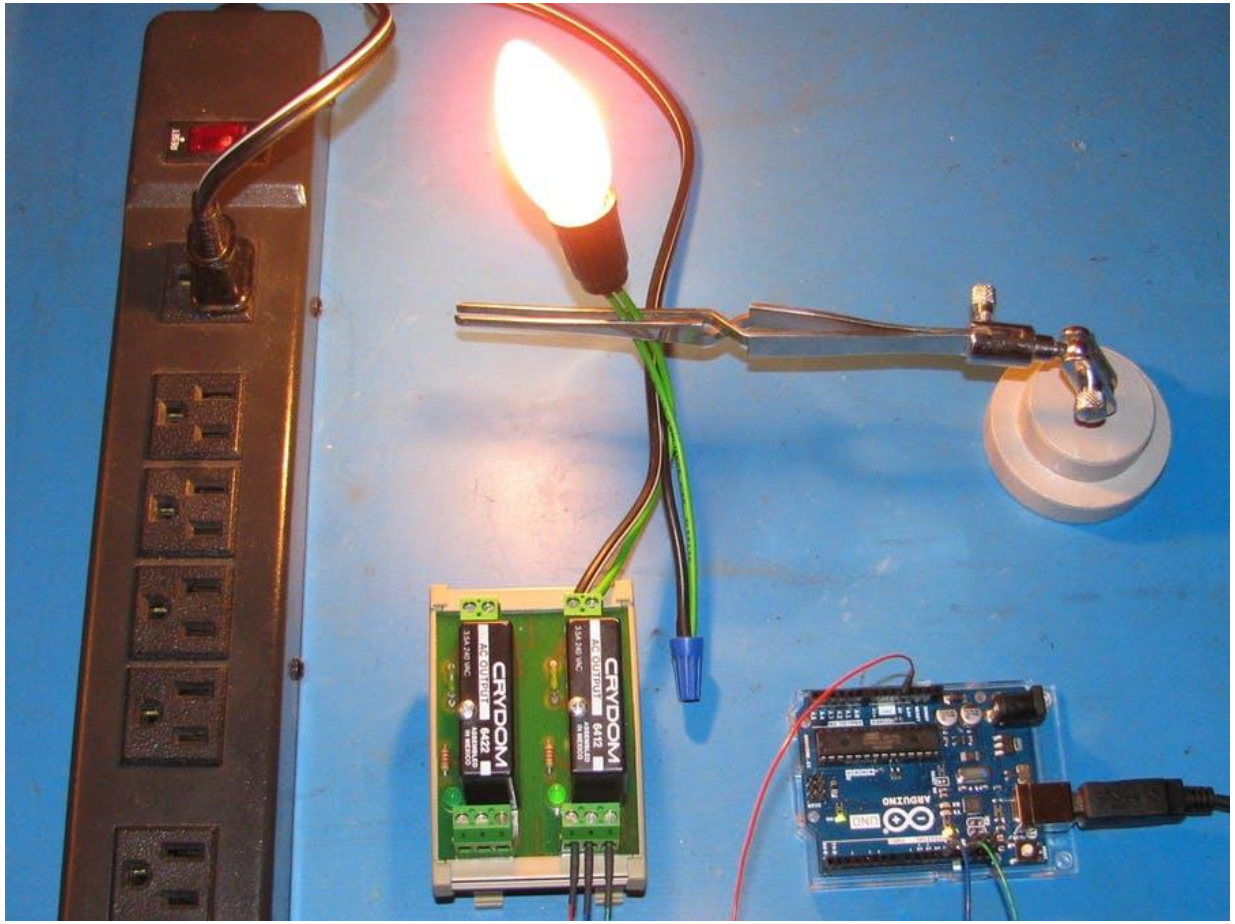
Arduino Shield





Arduino Project





Arduino Project

